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Generate random number from a specific probability mass in R

Asked 6 years, 5 months ago Modified 6 years, 5 months ago Viewed 1k times



2



How can I generate sample from a distribution with probability mass $P(X = x)$ in R? I know that probability mass, but it is not from a known distribution, also it is not linear, instead it has a complicated form. Can I use the inverse cdf method on the density, by working out the cdf and inverting it $X = F^{-1}(U)$?

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edited Apr 14, 2017 at 20:41

asked Apr 14, 2017 at 19:57



whuber ♦

313k

59

717

1239



Sedi

99

7

2 ▲ Use the `sample` function. With the argument `replace=TRUE` this simulates from the specified pmf using the alias method, see [related thread](#) – [Jarle Tufto](#) Apr 14, 2017 at 20:15

▲ @JarleTufto Can that approach be used if there are infinitely many x with $P(X = x) > 0$?
– [mark999](#) Apr 14, 2017 at 20:19 ✎

▲ @JarleTufto This assumes that X has finite support. – [nth](#) Apr 14, 2017 at 20:20

▲ The OPs suggested approach won't work because for discrete distributions the transformation is not 1-1. – [Michael R. Chernick](#) Apr 14, 2017 at 20:21

▲ The thread that Jarle Tufto linked indicates that the method is efficient when X has finite support. It works when there are infinite values as would be the case for the Poisson distribution but efficiency is not claimed in that case. – [Michael R. Chernick](#) Apr 14, 2017 at 20:27

|

2 Answers

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3



While it's possible to use methods based on the inverse-cdf if you don't know anything more than the probability mass function, but in many cases it can be terribly inefficient especially if it's unbounded. [If all you have is the pmf you can build a table of the cdf for say the bottom 99.9% of the distribution, for example, which will work but you still have to search it in some fashion -- e.g. binary search starting at the median might make sense, and you still have to do something else to deal with the tail beyond your table.]

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need to sample billions of times, or many people need to use this thing, then it would be worth investing the effort to make it very efficient).

There are many approaches that work directly with the pmf, including a number of table methods and accept-reject methods that might be suitable for your problem. There are several techniques described in my answer in [How to sample from a discrete distribution](#) but there's not enough information in your question to make concrete suggestions (such as whether one of those methods or some other might suit you better) -- it's hard to give explicit advice in the absence of details.

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edited Apr 15, 2017 at 1:43

answered Apr 14, 2017 at 23:37



Glen_b

275k

37

605

1005



I added some detail on accept-reject (rejection method) at the linked post – Glen_b Apr 19, 2017 at 4:02



3

Yes, you can use the inverse CDF method. Let $x_1, x_2, \dots$ be the (possibly infinite) sequence of elements which have positive probability masses $p_1, p_2, \dots > 0$. Let $I \in \mathbb{N}$ have distribution $\mathbb{P}(I = i) = p_i$. Generate a uniform random variable U and then let J be such that



$$\mathbb{P}(I < J) = F_I(J) \leq U < F_I(J + 1).$$



Take $X = x_J$. Then



$$\mathbb{P}(X = x_j) = \mathbb{P}(F_I(j) \leq U < F_I(j + 1)) = F_I(j + 1) - F_I(j) = p_j.$$

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edited Apr 14, 2017 at 21:02

answered Apr 14, 2017 at 20:44



nth

796

4

9